1. Research Problem

Clinical Gap

Short-term continuous glucose monitoring (CGM) enables reliable calibration of individualized insulin dosing parameters in adults with Type 1 Diabetes (T1D). However, after CGM discontinuation due to cost , burden or skin-related issues. Patients revert to static dosing calculations despite continued self-monitoring via finger-stick measurements. This frequently results in deterioration of glycemic control. This challenge is particularly pronounced in low- and middle-income countries (LMICs), where continuous CGM and insulin pump technologies are often financially inaccessible, leading to preventable long-term complications.

While physiological insulin and glucose models are well established, there is limited clinical evidence on whether machine learning (ML) and reinforcement learning (RL) based adaptive algorithms, initially calibrated using CGM, can maintain clinically acceptable outcomes under intermittent glucose monitoring, particularly when accounting for known circadian variation in insulin sensitivity.

Central Research Question

Can physiology-informed ML and RL insulin dosing algorithms calibrated during short-term CGM and incorporating circadian insulin sensitivity maintain clinically non-inferior glycemic outcomes using intermittent glucose measurements alone?

2. Research Objectives

Primary Objective

To clinically validate the non-inferiority of ML/RL-assisted insulin dosing algorithms using intermittent glucose measurements compared to continuous CGM-based management.

Primary Endpoint

- Change in HbA1c over 6 months (non-inferiority margin: +0.3%)

Secondary Endpoints

- Severe hypoglycemia (<0.5 events per patient-year)
- Time-in-range (70–180 mg/dL)
- Algorithm usability and interpretability (System Usability Scale)

Methodological Objectives

To develop and validate **physiology-informed ML and RL algorithms** for:

- Adaptive estimation of insulin sensitivity and correction factors, including circadian (time-of-day dependent) variation
- Learning individualized meal absorption dynamics from real-world data
- Estimating glucose trajectories between sparse self-measurements with quantified uncertainty

3. Methodology

Study Design

- Randomized controlled non-inferiority trial
- Duration: 6 months
- Sample size: 100 adults with Type 1 Diabetes

Study Arms

Arm A – ML/RL Algorithm + Intermittent Monitoring

- 2 weeks CGM-based calibration of patient-specific parameters
- 6 months finger-stick monitoring (3–4 measurements/day)
- Insulin dosing suggestions generated by adaptive ML/RL algorithms that update patient-specific parameters based on observed glycemic outcomes and circadian patterns

Arm B – ML/RL Algorithm + Continuous CGM

- Continuous CGM with identical ML/RL-assisted dosing logic, serving as the reference condition

This design isolates the effect of measurement density while holding the learning algorithms constant.

Key Inclusion Criteria

- Type 1 Diabetes ≥ 1 year
- Age 18–65
- Completed CGM calibration phase
- Baseline HbA1c 7.0–9.5%

Data Collected

- **Clinical:** HbA1c, hypoglycemia events, time-in-range
- **Algorithmic:** parameter convergence, circadian sensitivity profiles, prediction error, dosing accuracy
- **Patient-reported:** usability, perceived trust, diabetes-related burden

4. Algorithmic Innovations (Core Scientific Contribution)

1. Physiology-Guided Reinforcement Learning

Reinforcement learning is used to adapt insulin dosing parameters (e.g., insulin sensitivity, correction factors) within clinically safe bounds defined by physiological models and guidelines. Learning is constrained to ensure interpretability and patient safety.

Innovation: Early clinical validation of constrained RL for insulin dosing under intermittent glucose monitoring.

2. Circadian-Aware Insulin Sensitivity Modeling

Time-of-day–dependent modulation of insulin sensitivity and correction factors to capture circadian phenomena such as the dawn phenomenon and evening insulin resistance. Circadian features (clock time, sleep–wake cycles) are integrated into ML models and refined through RL-based adaptation.

Innovation: Clinical evaluation of circadian-aware insulin dosing suggestions rather than static 24-hour parameter assumptions.

3. Dual-Baseline Validation Framework

A methodological separation between:

- **Fixed-target glucose modeling** for clinical validation of algorithmic accuracy
- **Dynamic glucose estimation** for real-time insulin dosing decisions

This enables transparent evaluation of algorithm performance independent of patient behavior.

4. Sparse-Data Glucose Trajectory Learning

Machine learning models estimate glucose trajectories between finger-stick measurements using insulin-on-board, meal activity, and circadian context, with uncertainty-aware confidence scoring. **Innovation:** Learning glycemic patterns from sparse data rather than continuous sensing.

5. Learned Meal Absorption Dynamics

Data-driven refinement of multi-phase carbohydrate absorption models to distinguish delayed absorption from insufficient insulin dosing and reduce insulin stacking.

5. Expected Outcomes

Scientific Contributions

- First randomized clinical evidence for ML/RL-assisted insulin dosing without continuous CGM
- Validated framework for clinical evaluation of adaptive insulin dosing algorithms
- Peer-reviewed publications in diabetes technology, digital medicine, and ML-in-health journals

Clinical Relevance

- Maintained glycemic control with reduced monitoring burden
- Transparent, clinician-auditable learning algorithms

Health System Impact

- Reduced reliance on continuous sensing technologies
- Scalable methodological framework applicable across healthcare settings